

ECON 1020: Principles of Microeconomics

Workbook on English and Math Prerequisites

Recitations on September 7 and 9

This workbook covers three things you will need throughout the semester: (1) the English phrases that show up constantly in economics readings and lectures, (2) the basic high-school math this course leans on, and (3) the English vocabulary used to talk about that math. None of it is economics content itself. You will get that starting with the first lecture. Work through the practice problems as you go; a full answer key is available separately for your TA to check your work.

1 English for This Course

Economics readings and lectures lean on a small set of phrases over and over. Knowing them cold will save you from getting stuck on the sentence structure instead of the economics.

Phrase	Meaning	Example
assume / suppose	take something as true, for the sake of argument	Assume that the price of coffee is fixed.
in other words	restates the previous idea more simply	Demand fell; in other words, people bought less.
that is (i.e.)	clarifies the previous point precisely	The good is inferior, that is, demand falls as income rises.
for example (e.g.)	introduces an example	Some goods are necessities, for example, rice.
in terms of	specifies the measure or perspective used	Cost is measured in terms of dollars.
with respect to	concerning; in relation to	How does demand change with respect to price?
holding constant / all else equal	assuming every other factor stays the same	Holding income constant, a higher price lowers quantity demanded.
as opposed to	in contrast to	A shift in the curve, as opposed to a movement along it.
even though / although	introduces a contrast	Even though the price rose, demand stayed high.
given that / provided that	assuming a condition holds	Given that supply is fixed, price depends only on demand.
therefore / thus / hence	introduces a conclusion	Price rose; therefore, quantity demanded fell.
however / nevertheless	introduces a contrast with what was just said	Costs rose sharply. The firm, however, kept prices unchanged.
on the other hand	introduces an opposing point	Buyers want low prices; sellers, on the other hand, want high ones.
roughly / approximately	close to a value, not exact	Demand fell by roughly 10 percent.

Phrase	Meaning	Example
at least / at most	a lower or upper bound	The firm needs at least 100 units to break even.
on average	the typical value across many cases	On average, households spend 20 percent of income on food.
as a result / consequently	introduces an effect	Wages rose; as a result, production costs increased.
namely / that is to say	introduces a specific detail	One input is fixed, namely, the size of the factory.

Practice 1.1: Fill in the blank

Use one phrase from the table above to complete each sentence. Each phrase is used once.

- _____ the price of gasoline rises, some drivers keep driving the same amount.
- The firm's revenue increased; _____, it hired more workers.
- _____ all other prices stay the same, a lower price for coffee should raise the quantity sold.
- The two goods are complements, _____, people tend to buy them together.
- _____ a monopoly, a firm with many competitors cannot set its own price.

Practice 1.2: Combine the sentences

Combine each pair of short sentences into one, using the connecting phrase given in brackets.

- The firm's costs are fixed this month. The firm should focus on increasing sales. [*given that*]
- Prices rose sharply last year. Consumers kept buying about the same quantity. [*even though*]
- Income rises. People buy more restaurant meals and fewer instant noodles. [*as a result*]

2 Math Refresher

2.1 Percentages and Percentage Change

A **percentage** expresses a part of a whole out of 100:

$$\text{percentage} = \frac{\text{part}}{\text{whole}} \times 100\%.$$

A **percentage change** compares a new value to an old one:

$$\text{percentage change} = \frac{\text{new value} - \text{old value}}{\text{old value}} \times 100\%.$$

Worked example. A cup of coffee rises in price from \$80 to \$100.

$$\text{percentage change} = \frac{100 - 80}{80} \times 100\% = \frac{20}{80} \times 100\% = 25\%.$$

Worked example. Quantity sold falls from 200 units to 150 units.

$$\text{percentage change} = \frac{150 - 200}{200} \times 100\% = -25\%.$$

The negative sign means a decrease of 25 percent.

Practice 2.1

1. What is 15% of 240?
2. A ticket price rises from \$300 to \$360. What is the percentage change?
3. Enrollment falls from 500 students to 425 students. What is the percentage change?
4. A rent of \$20,000 per month increases by 8%. What is the new rent?
5. If a value decreases by 20% from 500, what is the new value?

2.2 Solving for a Variable

Many formulas in this course relate two or more variables. You will often need to isolate one of them. The rule is simple: whatever you do to one side of an equation, you must do to the other side too.

Worked example. Suppose quantity demanded is $Q_d = 100 - 2P$, and we are told $Q_d = 40$. Solve for P .

$$40 = 100 - 2P \quad \Rightarrow \quad 2P = 100 - 40 = 60 \quad \Rightarrow \quad P = 30.$$

Worked example. Solve $y = mx + b$ for x .

$$y - b = mx \quad \Rightarrow \quad x = \frac{y - b}{m}.$$

Practice 2.2

1. If $Q_d = 120 - 3P$ and $Q_d = 60$, solve for P .
2. If $Q_s = 4P - 10$ and $Q_s = 30$, solve for P .
3. Solve $P = 50 - 0.5Q$ for Q .
4. Solve $A = \pi r^2$ for r .

2.3 Slope and Linear Equations

The **slope** of a line measures how steep it is: how much y changes for every one-unit change in x .

$$\text{slope} = \frac{\text{rise}}{\text{run}} = \frac{y_2 - y_1}{x_2 - x_1}.$$

A positive slope rises from left to right; a negative slope falls from left to right; a slope of zero is a flat, horizontal line.

The general equation of a line is $y = mx + b$, where m is the slope and b is the **y -intercept**, the value of y when $x = 0$.

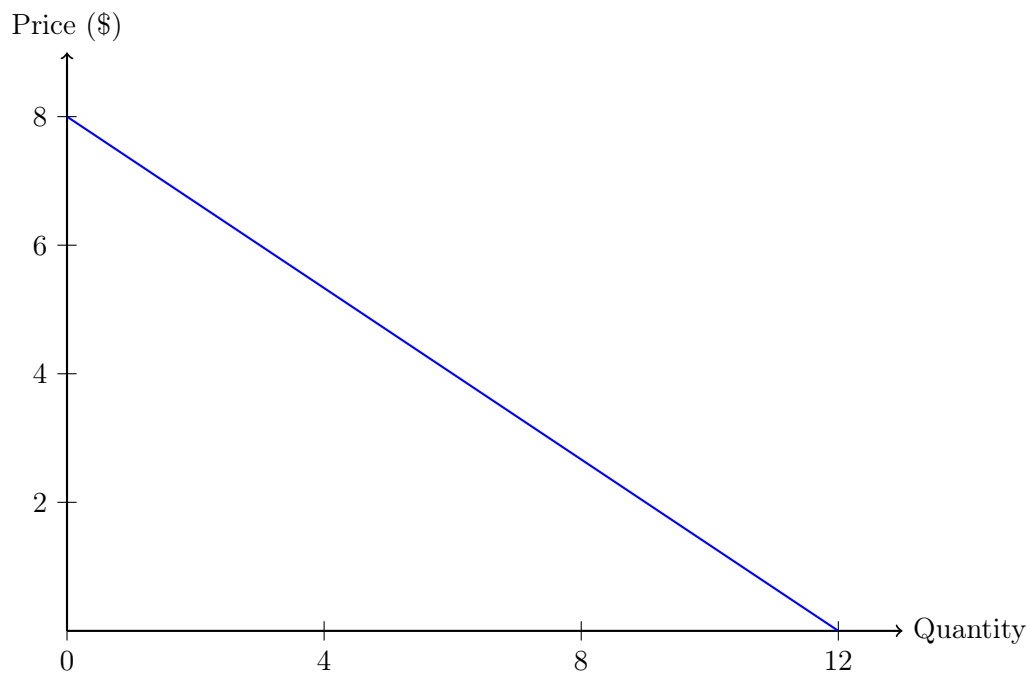
Worked example. A line passes through $(0, 10)$ and $(5, 0)$.

$$\text{slope} = \frac{0 - 10}{5 - 0} = -2, \quad y\text{-intercept} = 10, \quad \text{equation: } y = -2x + 10.$$

Practice 2.3

1. Find the slope of the line through $(0, 20)$ and $(4, 12)$.
2. Find the slope of the line through $(2, 5)$ and $(6, 5)$. What does this slope tell you about the line?
3. A line has a y -intercept of 30 and a slope of -3 . Write its equation. What is the value of y when $x = 6$?
4. Is a steeper line's slope larger or smaller in absolute value than a flatter line's slope?

2.4 Reading a Graph



The line above shows how price and quantity relate to each other for some good.

Practice 2.4

1. What is the y -intercept of this line? What does that point mean here?
2. What is the x -intercept of this line?
3. What is the slope of this line?
4. If quantity increases by 6, what happens to price?
5. Is this an example of a straight line or a curve? Would you call the slope positive or negative?

3 Math Terms in English

This section is the overlap between Sections 1 and 2: the specific English words you need to talk about graphs, equations, and numbers out loud in class.

Term	Meaning	Example
Cartesian plane (co-ordinate plane)	the flat surface on which a graph is drawn, defined by a horizontal and a vertical axis	Every point in the coordinate plane has an x -value and a y -value.
axis (plural: axes)	one of the reference lines on a graph	Quantity is measured on the horizontal axis.
origin	the point $(0,0)$ where the axes meet	The line does not pass through the origin.
quadrant	one of the four regions the axes divide the plane into	This graph only uses the quadrant where both x and y are positive.
intercept	the point where a line crosses an axis	The y -intercept is 10.
slope	the steepness of a line; rise over run	The demand curve has a negative slope.
positive relationship	a relationship in which the two variables move in the same direction	Income and spending have a positive relationship.
negative relationship	a relationship in which the two variables move in opposite directions	Price and quantity demanded have a negative relationship.
coordinate	a pair of numbers (x,y) that locates a point	The point has coordinates $(4,8)$.
to plot (a point)	to mark a point's location on a graph	Plot the point $(4,8)$ on the graph.
curve	a line on a graph, straight or bent, showing a relationship	The supply curve slopes upward.
linear	describes a relationship that graphs as a straight line	This is a linear equation.
variable	a quantity that can take different values	Price and quantity are both variables here.
constant	a fixed value that does not change	The 10 in $y = 2x + 10$ is a constant.
equation	a statement that two expressions are equal	Write an equation for this line.
ratio	a comparison of two quantities by division	The ratio of price to quantity is 5 to 1.
proportion	a statement that two ratios are equal	These two variables are in proportion.
rate	a quantity measured against another quantity	The growth rate was 3 percent.
rate of change	how quickly one variable changes as another does	The slope is the rate of change of y with respect to x .
marginal	describing the effect of one additional unit	The marginal cost is the cost of producing one more unit.
to increase/decrease by __%	to change by a given percentage	Demand decreased by 10 percent.

Term	Meaning	Example
percentage point	a unit of difference between two percentages (not the same as “percent”)	If unemployment goes from 5% to 7%, it rose by 2 percentage points, not 2 percent.
steep / flat	describes a large or small slope	The steep part of the curve is on the left.
to shift (a curve)	to move an entire curve to a new position	An increase in income shifts the demand curve.
movement along (a curve)	to move from one point to another on the same curve	A price change causes movement along the demand curve.
independent variable	the input variable, usually plotted on the x -axis	Price is often the independent variable.
dependent variable	the output variable, usually plotted on the y -axis	Quantity demanded is the dependent variable.
to solve for	to find the value of an unknown	Solve for P in this equation.
to substitute	to replace a variable with a known value	Substitute $P = 20$ into the equation.
to rearrange	to change the form of an equation	Rearrange the equation to isolate Q .
trade-off	giving up one thing to get another	Every choice involves a trade-off between alternatives.
ceteris paribus (all else equal)	assuming everything else stays the same while one variable changes	Ceteris paribus, a higher price leads to lower quantity demanded.

Practice 3.1: Describe the graph

Using the graph from Section 2.4 and at least five terms from the glossary above, write three complete English sentences describing it.

Reading Math Expressions Out Loud

Class discussion often means reading an expression off the board out loud, and a few conventions make this easier once you know them.

Expression	How to say it out loud
$P = 25$	" P equals 25," or " P is equal to 25."
$P \neq 25$	" P does not equal 25," or " P is not equal to 25."
$P > 25$	" P is greater than 25."
$P < 25$	" P is less than 25."
$P \geq 25$	" P is greater than or equal to 25."
$P \leq 25$	" P is less than or equal to 25."
$P \approx 25$	" P is approximately 25," or " P is about 25."
ΔP	"The change in P ."
P_1, P_2	" P sub one" or " P one"; " P sub two" or " P two."
$2x$	"Two x ." (No need to say "times.")
$\frac{\Delta Q}{\Delta P}$	"Delta Q over delta P ."
10%	"Ten percent."
x^2	" x squared."
x^3	" x cubed."
x^n	" x to the n -th power," or " x to the n ."
P^1, P^2	Not an exponent: a superscript label for a category or scenario, read as " P superscript one" or, once the context is clear, just " P one."

Practice 3.2: Say it out loud

Write out, in words, how you would say each of the following out loud in English.

1. $y = 2x + 3$

2. ΔP

3. P_1

4. $Q \geq 40$

5. $P \approx 25$